IRAI 2026
IEEE International Conference on Responsible Artificial Intelligence

Generative Scenario Modeling for Robust Dispatch in Renewable-Integrated Smart Grids

Sean Afamefuna · Sadia Rahman · Glory Simon · Jessica Osunbor
University of Vaasa, Finland · Halmstad University, Sweden · Shenzhen University, China

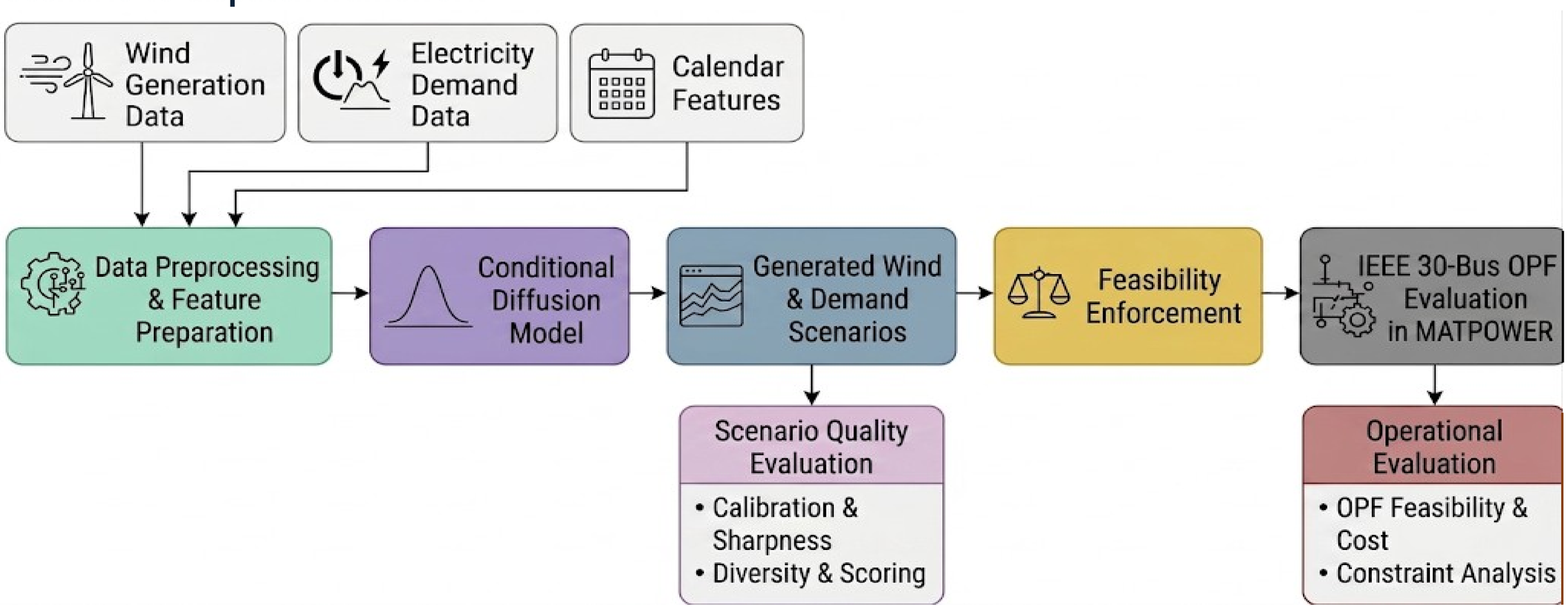
1 Motivation and objective

- Renewable-rich grids face short-term wind and demand uncertainty.
- Single forecasts can miss ramps and high-load operating extremes.
- We generate usable 24-hour scenarios and test their dispatch impact.

2 Main contributions

- Conditional diffusion: 48-hour history -> joint 24-hour wind-demand scenarios.
- Feasibility layer: non-negativity + data-driven ramp correction.
- End-to-end test: 24,000 IEEE 30-bus AC OPF runs in MATPOWER.

3 Scenario-to-dispatch framework



Joint wind-demand scenarios are generated, feasibility-corrected, and evaluated through probabilistic scores and IEEE 30-bus OPF.

4 Probabilistic scenario quality

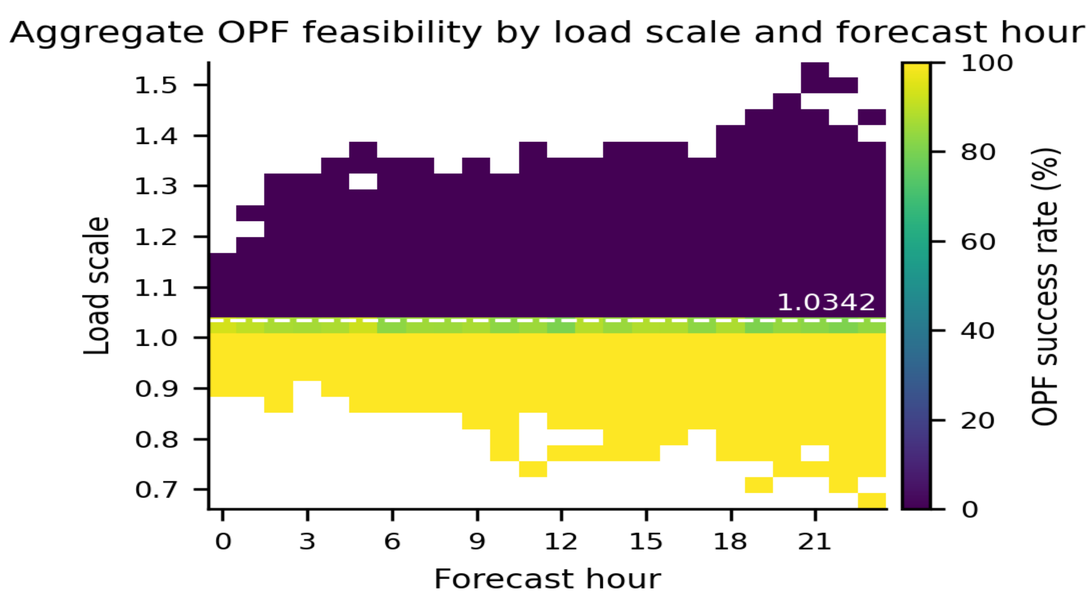
Probabilistic Scenario Quality – Test Set

Lower values are better.

Method	Wind CRPS	Load CRPS	Energy Score
Persistence	5804.28	9149.97	65116.32
Linear Regression	**5015.03**	3189.96	36190.77
Historical Bootstrap	7441.67	5461.63	51456.35
Diffusion	5231.56	**1974.40**	**32432.48**

Diffusion achieves the strongest load and joint probabilistic performance.

5 Scenario-driven OPF stress test



Yellow = high OPF success; purple = low success; dashed line = ~1.0342x load-scale boundary.

6 Feasibility correction and key findings

Feasibility Correction — Test Set

	Raw diffusion output	→	After feasibility correction
Wind ramp violations	2,597	→	0
Load ramp violations	115	→	0
Negative wind entries	232	→	0
Energy Score	32,432.48	→	32,293.74

All observed violations were eliminated without degrading probabilistic quality.

24,000
scenario-hour OPF runs

59.19%
overall OPF success rate

1.0342x
approx. feasibility boundary

565.80
mean successful OPF cost

- **Generated scenarios reveal the case30 feasibility boundary, not just forecast error.**
- **Feasibility correction removes observed violations while preserving scenario quality.**